# JINGTIAN WU

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## EDUCATION

**Cornell University** *May 2026 (expected)*
B.A. in Computer Science, Minor in Artificial Intelligence
GPA: 3.97

## RESEARCH INTERESTS

Natural Language Processing, Multimodality, Machine Learning, Reinforcement Learning

## PUBLICATIONS

[1] **Jingtian Wu** and Y. Hicke, *Joint Training Optimization: Calibrating Paired LLM Personas via Self-Play*, Accepted (Poster), PersonaLLM: Workshop on LLM Persona Modeling @ NeurIPS 2025.
[2] **Jingtian Wu**, J. Guo, G. Guo, and Q. Yang, *"The AI is Organic": Fostering Music Ideation with Multimodal AI Inspirations*, Under review (Invited for Revise & Resubmit), CHI 2026.
[3] **Jingtian Wu** and C. Cardie, *Reasoning court: Combining reasoning, action, and judgment for multi-hop reasoning*, 2025. arXiv: `2504.09781 [cs.CL]`. [Online]. Available: `https://arxiv.org/abs/2504.09781`.
[4] **Jingtian Wu**, "Algorithmic composition of music utilizing the digits of pi," in Highlights in Science, Engineering and Technology, vol. 85, 2024, pp. 570–577.
[5] X. Zhao, Y. Zhou, **Jingtian Wu**, Q. Xu, and Y. Zhang, "Turn real people into anime cartoonization," in Proc. IEEE ICCECE, 2021, pp. 387–393. DOI: `10.1109/ICCECE51280.2021.9342433`.

## RESEARCH EXPERIENCE

**LLM Persona Modeling: Behavioral Alignment in AI-Simulated Personas** Summer 2025 – Present
*Cornell NLP Group | Advisor: Claire Cardie; Collaborator: Yann Hicke* Cornell University

- Proposed a **self-play reinforcement learning** framework that jointly trains paired LLM personas to co-adapt and align their behaviors toward a shared target [1].
- Evaluated the method in AI-simulated doctor-patient dialogues, demonstrating data-free behavioral calibration and laying the groundwork for human-like persona simulations in general education and training contexts.
- Currently extending the framework to multi-turn interactions and domain-agnostic settings.

**OmniWizz – A Multimodal Music Ideation Assistant** Summer 2025 – Present
*Advisors: Qi Yang and Ge Guo* Cornell University

- Co-led the design and development of OmniWizz [2], an interactive **multimodal system** generating coordinated text, image, and audio outputs to inspire music creation.
- Independently built the full-stack architecture enabling flexible cross-modal input–output workflows that support diverse and innovative creative exploration.
- Conducted a 16-participant study combining quantitative and qualitative analyses, showing OmniWizz expanded ideation, reduced fixation, and preserved creators' sense of ownership. [View Demo]

**Automatic Synthetic Data Generation** Spring 2025
*Cornell NLP Group | Advisor: Claire Cardie* Cornell University

- Exploring automated dataset construction for **self-evolving models**, developing a teacher–student LLM framework inspired by model distillation and curriculum learning, where an expert LLM acts as an in-environment teacher to generate domain-specific questions, decompose tasks, and refine the student model through guided interaction for scalable continual adaptation in medicine.
- Built a hierarchical knowledge map expansion pipeline, covering 226 subfields, generating 2.9K+ teacher–student dialogues and MCQA datasets without human annotation.
- Fine-tuned the student LLM, achieving +6.76% accuracy over baseline with 6.4× fewer training samples.

**Culturally and Temporally Contextualized Lyric Generation** Spring 2025
*Advisor: Matthew Wilkens* Cornell University

- Investigating how to adapt LLMs for lyric generation that is culturally and temporally grounded.
- Curated a 31,000-song corpus and fine-tuned LLaMA 3.1 8B with LoRA for context-aware generation.
- The resulting model achieved +3.4% CoSENT, +67.4% ROUGE-2, and +1.6% BERTScore improvements, with human evaluators rating its cultural and temporal relevance 36% higher on average.

**Automatic Code Generation System** Fall 2024
*Cornell NLP Group | Advisors: Claire Cardie and Wenting Zhao* Cornell University

- Extended Aider with a multi-agent system to explore how LLMs can function more effectively as autonomous software engineers
- Designed components for task planning, debugging, and progress tracking, orchestrated through an automatic iteration loop. Proposed safeguards including a Git-based rollback mechanism and iterative refinement module, preserving best-performing function versions.
- On the Commit-0 Parsel benchmark, the system boosted accuracy by 15.5 percentage points over baseline.

**Enhancing Multi-Hop Reasoning in LLMs** Summer 2024
*Cornell NLP Group | Advisors: Claire Cardie and Wenting Zhao* Cornell University

- BURE Researcher; awarded $7,000 fund for research in Summer 2024.
- Tackled LLM hallucinations in multi-hop QA by developing Reasoning Court [3], extending ReAct-style reasoning–retrieval with two agents and an LLM judge that selects or synthesizes factually grounded answers.
- Outperformed few-shot prompting baselines on HotpotQA, FEVER, and MuSiQue, achieving up to +7.2% EM and +10.5% F1 improvements while reducing LLM calls compared to hybrid baselines.

**Computer Music Synthesis and Composition** Summer 2023
*Advisor: Roger B. Dannenberg* Carnegie Mellon University

- Explored mapping $\pi$'s digits to the minor pentatonic scale, using programming to simulate traditional Chinese instruments for mathematically grounded music.
- Designed an algorithmic method to generate a melody and an automated countermelody.
- Composed an original piece and showcased the creative potential of algorithmic music composition.

**Algorithms for Big Data** Summer 2021
*Advisor: David Woodruff* Carnegie Mellon University

- Tackled inefficiency of GAN-based cartoonization, aiming for lightweight, high-quality anime–style results suitable for resource-constrained devices.
- Sole developer of *LWAnimeGAN* by combining AnimeGAN and photo2cartoonGAN with architectural optimizations—reduced output channels, depthwise separable convolutions, and removal of HourGlass blocks.
- Reduced parameters by 8.8%, FLOPs by 56.1%, and inference time by 18.6% while preserving visual fidelity.

## RESEARCH-RELATED PROJECTS

**LLM Framework for Academic Peer Review** Spring 2024
*Advisor: Thorsten Joachims; Collaborator: Zhaolin Gao* Cornell University

- Led full-stack development of the demonstration system for the research paper *Reviewer2: Optimizing Review Generation Through Prompt Generation*.
- Contributed to the NSF-supported *Frameworks: arXiv as an Accessible Large-Scale Open Research Platform*.
- Presented the system at BOOM 2024 (Bits On Our Minds), featured in the *Cornell Chronicle*.

## AWARDS & HONORS

**Honorable Mention, CRA Outstanding Undergraduate Researcher Award** 2026
Annual national recognition awarded to the top ≈200 undergraduate students in North American universities who show outstanding research potential in an area of computing research. Nominated by Cornell University as one

of 2–4 students from the Department of Computer Science.

**Dean's List of the College of Arts and Sciences** Fall 2022, Spring 2024

## TEACHING EXPERIENCE

**CS 3700: Foundations of AI Reasoning and Decision-Making, Cornell** Fall 2024, Fall 2025

– Led weekly office hours and guided students through complex topics (e.g., heuristic search, game-play, and reinforcement learning), provided exam preparation support, and responded to questions on Ed discussion.

## SKILLS

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| **Languages** | Python, JavaScript, Java, LaTeX, Markdown, HTML/CSS |
| **Machine Learning / NLP** | PyTorch, Hugging Face Transformers, LoRA/QLoRA, Unsloth |
| **Software Development** | FastAPI, Flask, Node.js, React |
| **Tools** | Git, Jupyter Notebook, Visual Studio Code |

## HOBBIES & INTERESTS

**Competitive Chess:** Tennessee State Champion; 4th place, U.S. National Blitz Championship; multiple-time Hunan Provincial Champion (China); Team Leader for Hunan Province, National Mind Sports Games (China); represented Cornell in the Inter-Ivy Chess Championship; peak Chess.com ratings of 2974 (bullet) and 2674 (blitz).

**Music Production:** Compose, arrange, and produce original and cover works across multiple genres; 2M+ total views and 400K+ likes on TikTok and Bilibili; 4th place, Cornell CSSA Singing Competition; former University Growth Specialist at Suno, leading campus ambassador outreach for AI music platform adoption.